Online Appendix: [From Experts to Language Models: Enhancing Online Expert Elicitation with LLMs]

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1 Document Preamble

This document is identified as the **Online Appendix 2**. It contains the coded discussions from the two expert elicitation sessions. Although these discussions are not part of the main paper, they are included in the repository to support understanding of the elicitation process and the topics raised.

2 Coding of participants

Table A1. Expertise - First Elicitation Study

S/No	Code	Expertise
1	S_1E1	DC power electronics
2	S_1E2	DC power electronics
3	S_1E3	DC power electronics
4	S_1E4	DC power electronics
5	S_1E5	Data Center electrification
6	S_1E6	DC protections
7	S_1E7	Ports electrification
8	S_1E8	DC power electronics
9	S_1E9	DC power electronics
10	S_1E10	Ports electrification
11	S_1E11	DC power electronics
12	S_1E12	DC protections
13	S_1E13	Academician

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Table A2. Expertise - Second Elicitation Study

S/No	Code	Expertise
1	S_2E1	DC power electronics
2	S_2E2	DC interoperability
3	S_2E3	DC power electronics
4	S_2E4	Buildings electrification
5	S_2E5	Academician
6	S_2E6	Academician
7	S_2E7	DC power electronics
8	S_2E8	DC power electronics
9	S_2E9	Academician
10	S_2E10	DC power electronics
11	S_2E11	AC/DC power systems

3 Discussion Per question

This section presents the discussions each expert raised on a particular question. The expert identifiers are provided in Table A1 and Table A2 for the first and second studies, respectively. The question texts are available in the EE protocols for both sessions, that is, sessions one and two, which are shared via the GitHub repository.

3.1 First EE Study

Feasibility question—first EE study. No discussion was captured.

Importance question—first EE study. No raw discussion captured. But since the smart and sustainable DC cables had concerns with response variations, the following was noted: The elicitation team clarified that the smart and sustainable DC cables aim to achieve an optimal cable design that minimizes the use of environmentally friendly raw materials.

Time for Widespread Adoption - First EE Study. The following were the coded discussions in regard to time for widespread adoption: Participant $\mathbf{S_1E_3}$ commented: *"Maybe it needs to be clarified at the beginning of the presentation what exactly a static protection system is. I would suggest including an example that shows where, in the design process, this system or its methodology is applied."*

Another concern related to the interpretation of "widespread" adoption. Participant $\mathbf{S_1E_7}$ noted: *"You have to take into consideration what you consider 'widespread.' If it's for investment purposes, or for the general public."* The elicitation team clarified that 'widespread' referred to adoption by the general public. This feedback was incorporated to improve the design of the second EE study protocol.

Risk of Delay in Adoption - First EE Study. The Direct Current (DC) cables being rated as having an insignificant but also critical risk, the following discussion was raised by experts concerning the relationship between conductor materials used for the DC cables and its impact on the cost. $\mathbf{S_1E_7}$ highlighted that: *"I put it in the critical range, because we are in a big project here. Doing the electrification of the ports, though it's not in DC, but we have about seven kilometers of cable to put in, it's a giant cost."* The concern was clarified by $\mathbf{S_1E_3}$, as: *"It is not only about the length of the cable* *in different applications or different installations. But, the point also is to check what is possible in terms of copper, or aluminum, because the cost will be higher according to the kind of conductor materials."*

Experts explained that the perception of lower or higher risk was largely due to the availability of alternative solutions already on the market. They suggested that, as long as existing alternatives meet functional needs, stakeholders would be unlikely to prioritize adopting newer technologies.

However, Participant $\mathbf{S_1E_3}$ challenged this view, emphasizing that the innovations embedded within the proposed DC solutions, such as intertripping wires and the use of optimized, sustainable materials for the smart and sustainable DC cables, were not fully understood by the panel. Furthermore, Participant $\mathbf{S_1E_2}$ argued that the adoption of sustainable cables would be greatly accelerated if regulations mandating their use were established.

Barriers to Adoption - First EE Study. During the discussion, the high ranking of the lack of trained personnel was highlighted. Participant $\mathbf{S_1E_{13}}$ noted: *"It's very interesting. It's very often disregarded in the literature, but in one of the first meetings we had for this project, this topic was raised even more than costs."* Also, added: *"I believe this result stems from the fact that we are experts; if we surveyed end users, the answers might differ significantly."*

Regarding the high ranking of high costs of DC solutions, Participant $\mathbf{S_1E_5}$ explained: *"I chose high costs as a barrier because the adoption of DC technology will depend on early adopters building demonstration sites without full standardization. For that to happen, you need a viable cost-benefit analysis, but the current prices of DC products—still not produced at scale—hinder this."*

Experts were also invited to suggest additional barriers. New barriers elicited from most to least perceived, included: lack of general public awareness, DC equipment interaction issues, lack of clear end-user advantages, unclear incentives, challenges with Total Cost of Ownership (TCO) comparisons, and the need for new technology.

On further inquiry about the "need for new technology" as a proposed barrier by an expert, Participant $\mathbf{S_1E_6}$ explained: *"We're moving into new protection technologies based on power electronics. This shift away from decades-old solutions might cause hesitation among stakeholders."*

Regarding the meaning of TCO comparison, Participant $\mathbf{S_1E_{10}}$ elaborated: *"TCO refers to comparing two solutions across a comprehensive scenario, covering not just installation costs but also operating costs and other factors over a longer period."* Participant $\mathbf{S_1E_3}$ agreed, adding: *"Users often compare technologies that are not at the same maturity level. The DC microgrid advantage should be compared against the AC grid standard, not against traditional AC installations only."*

Moreover, Participant $\mathbf{S_1E_4}$ shared a practical example illustrating the challenge of public understanding: in one project, although DC was technically suitable for a building, end-users did not perceive a clear advantage over conventional AC systems and therefore showed little interest.

Concerning the application of DC in demonstration projects, particularly in ports, Participant $\mathbf{S_1E_{13}}$ reflected: *"In some demos, it feels more obvious. For example, data centers clearly benefit from DC. But for ports, the roadmap is less clear. We're discussing onshore power supply while already introducing DC—it's like trying to leap before learning to walk."*

Open Discussion - First EE Study. During the open discussion, experts were invited to suggest any additional DC solutions they considered vital based on the targeted demonstrations.

Participant $\mathbf{S_1E_2}$ emphasized the importance of developing an Energy Management System (EMS) tailored specifically for DC applications, emphasizing that, while the development of individual devices is important, an interoperable EMS is crucial for ensuring the coordination and integration of different equipment. Establishing a common platform at the EMS level was seen as a key enabler for broader implementation of DC systems.

Building on this point, Participant $\mathbf{S_1E_{13}}$ highlighted related developments within the project, explaining, alongside efforts to create digital twins, the EMS has emerged as a particularly valuable tool, as it supports system management and interoperability across DC technologies. Also, added: *"We are also developing digital twins. But even beyond that, the EMS is something we already have within the project, and it's particularly interesting because it acts as one of the core tools enabling interoperability."*

3.2 Second EE Study

This part presents the discussion coding of the second EE study.

Feasibility and Importance - Second EE Study. Following this, the elicitation team gathered expert insights on the importance of each DC solution. The results are summarized in **??**, covering all five (5) solutions. For the smart and sustainable DC cables, five (5) participants rated them as "important," another five (5) as "very important," and one (1) as "not important." The elicitation team requested clarification regarding the "not important" rating. Participant $\mathbf{S_2E_8}$ explained: *"It's not that we don't need the DC cables to be smart. My point was that it's not essential. We need cables, but there are already cables, and we've found that you typically shouldn't need extra insulation with them. So, in some cases, you could use Alternating Current (AC) cables if you wanted to."*

Regarding the DC connectors, six (6) experts rated as "very important," while one (1) rated them as "not important." Experts raised concerns that connectors have already been extensively researched, suggesting that existing solutions may suffice. They further noted that not all grid components are designed for plug-and-play installations. Participant $\mathbf{S_2E_2}$ elaborated: *"Not everything is pluggable. If you're doing an AC installation from the distribution cabinet, nothing is plugged. Depends on where you are."* Explained, the necessity of DC connectors depends on the location within the grid.

For the solid-state circuit breaker, nine (9) experts rated as "very important", one (1) as "important", while one (1) rated it as "not important." Although no direct explanation was provided for the "not important" rating, participant $\mathbf{S_2E_6}$ emphasized the critical role of this DC solution: *"Actually, if we are thinking about interrupting currents in DC, it's a big deal. So, these devices must be important. We have to put some thinking when building them because they will be much harder to build than the other ones in AC."*

Other DC solution experts rated the LVAC-LVDC interlink converter. Having nine (9) ratings as "very important", and two (2) as "somewhat important". Among the concerns raised is from participant $\mathbf{S_2E_3}$ explaining: *"That would not be the point is that we think as a Europeans. So we think of adding something to an existing AC network. It's linked with our existing electrical network."* This was further elaborated, being concerned with the context of application in the DC grids. On the other hand, a concern on high voltage DC interlink was raised by Participant $\mathbf{S_2E_6}$ discussing: *"Maybe we should think about high voltage DC interlinks, because if we are thinking about maritime technologies, it's easy to have 10 megawatt. If you are talking about 400 volts, the currents will exponentially increase. So high voltage will be something to consider."* This was agreed by participant $\mathbf{S_2E_3}$ adding: *"Your point is relevant, though we are not talking about high voltage."* The input was supported by other experts on the need to consider the future of port electrification separately, given the uncertainties related.

For the network design tool for DC solutions, eight (8) experts rated it as "very important", and three (3) as "important". Though the ratings were positive, the elicitation team sought insights from experts on this solution. Participant $\mathbf{S_2E_3}$ explained: *"It's much too complicated as there is a possibility to get different sources in the same network and so on. It seems very complicated to organize and to define a network without a software tool."* This claim received agreement from participant $\mathbf{S_2E_7}$: *"The more the complicated microgrid, the more the simulation tools are required and second thing the* *robustness of the system also defines the optimization options that opens up for a second thing. It's a cost-sensitive matter on solutions' target countries and where they are developed."* The elicitation team supported the contributions, explaining its alignment with the project.

Barriers to Adoption - Second EE Study. The elicitation team explained the efforts to address incompatibility that are currently explored by Open Direct Current Alliance (ODCA) and Current-OS. They also inquired for insights from experts. Participant $\mathbf{S_2E_4}$ added: *"Though Current-OS and ODCA can certainly help, but we will need much more to have enough trained personnel in DC systems. The universities and different bodies that can deliver training services will need to contribute. So, ODCA and Current-OS can deliver a frame of courses or a book or a backbone or assets, and content, but probably not be capable to make and deliver those training services."* The contribution was supported by participant $\mathbf{S_2E_1}$: *"We have the expertise, but we are not having training or certification bodies to provide that kind of training to skilled people or universities."*

To deepen the discussion, participants were invited to suggest any additional barriers they considered critical but not previously listed. From most to the least cited additional barriers included: the lack of a DC grid code, high , limited test facilities for DC systems, regulatory gaps, resistance to change, inertia in mindset shifts, limited understanding of DC benefits by public authorities, robustness issues, and the aging of power converters. Fault protection in DC systems was emphasized separately under two labels: "DC fault protection" and "fault protection and mitigation." Investor preferences and interests were also identified as potential barriers.

Additionally, participant $\mathbf{S_2E_4}$ contributed: *"If the public does not support, and they cannot support if they don't understand, then the grid code has little chance of changing. I'm not sure that the utilities have the will to move forward, as they are AC companies by history. So, we need the support of the public authorities. We are already working at the European Commission on the set plan for DC technology."*

On the other hand, participant $\mathbf{S_2E_1}$ explained: *"General unwillingness to change in our societies, DC is a different way than it has been done before. If a person is not willing to change, they will not even read a regulation."* The participant also added: *"Like within the EU Fit for 55 or Green Deal initiatives, that puts a lot more emphasis on efficiency, and that is the driver, and then it was just the two different conferences were also DC was a highlight. These third industry players are the head of the regulatory or general public about the need to decarbonize, and energy efficiency is left at the center, and DC plays an important role."* The discussion concluded by highlighting the necessity of securing strong support from public authorities, as the European Commission is already pursuing strategic efforts.

Open Discussion - Second EE Study. The open discussion in the second elicitation study raised several additional critical points.

One major topic was the urgent need to train personnel specifically for DC systems, particularly at the technician level, adding to what was already discussed in **??**. Participant $\mathbf{S_2E_2}$ emphasized: *"We need education and don't think of engineers, we need technicians to train practical people."*

Also, participant $\mathbf{S_2E_6}$ stressed on co-existence between AC and DC infrastructures rather than favoring one over the other. Quoted: *"At present, all the loads are AC mostly. So if we are thinking in a change from AC to DC, we should consider a point where they both coexist. Because if you want to change it directly, we have to always install something in the middle of what we have already and something that we should add extra exactly to put the things we already have working and that will be always an extra price."*

In contrast to participant $\mathbf{S_2E_6}$ insights, participant $\mathbf{S_2E_2}$ argued: *"It's about installing loads of renewable including storage systems, electrolyzers, and the like, which are all DC systems. Connecting loads of DC to AC grids. It's not changing* *the AC grids."* The participant added: *"Doing DC, you need to start from scratch because it's completely different, due to difference in physical principles."* Interestingly, participant $\mathbf{S_2E_5}$ admitted: *"Completely agree, that is the biggest truth we heard today."*

Experts were also invited to suggest additional DC solutions relevant to the project's planned demonstrations. Participant $\mathbf{S_2E_7}$ responded: *"All the technologies are already in place. We have only to bring awareness on the DC systems to the general public."*